

Design of Local Area DGNSS Architecture to Support Unmanned Aerial Vehicle Networks: Concept of Operations and Safety Requirements Validation

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ABSTRACT

A local-area differential global navigation satellite system (LAD-GNSS) provides high integrity and accuracy for unmanned aerial vehicles (UAVs). In this study, three major issues were investigated to fully establish the LAD-GNSS and analyze the performance of the system. First, UAV operation concept and requirements are defined. This includes segregation of UAV operation coverage airspaces, and derivation of navigation requirements, alert limits (ALs), for each operational coverage. Second, LAD-GNSS architecture was designed by simplifying hardware and monitoring algorithms both on a ground facility and onboard module using the well-established GBAS as a starting point. Lastly, theoretical performance evaluations were performed by comparing position uncertainty bounds, which are represented by protection levels (PLs), to the corresponding navigation requirements (ALs) for each coverage. ALs and PLs under both nominal and malfunction cases are derived and compared correspondingly. In addition, a method is described for deriving PLs for excessive acceleration and code-carrier divergence fault scenarios, which are bounded by using a maximum-allowable error in range. Using these PLs, the integrity/continuity allocations are ideally and dynamically assigned to each single-fault hypothesis to obtain optimized PLs that are identical for all fault scenarios. The results show that the VPLs are reduced by approximately 16% when implementing the optimal allocation method.

1. INTRODUCTION

One of the most essential technologies required for the operation of unmanned aerial vehicles (UAVs) is reliable and accurate computation of the navigation solution. The concept of a local-area UAV network that utilizes local-area differential global navigation satellite system (LAD-GNSS) solutions was proposed to achieve high navigation accuracy and reliability, which are required for autonomous UAV applications [1]. LAD-GNSS meets the required level of integrity requirements (comparable to those of GBAS Category I–III operations) by monitoring navigation faults at both the ground station and UAV and by broadcasting integrity information to the UAV. By utilizing the integrity information, which defines

navigation system error models, the UAVs can compute their conservative position error bounds (i.e., protection levels (PLs)) and consequently safe separation distances in real time. The LAD-GNSS architecture for the local-area UAV network proposed in previous work includes several methods of simplifying the current GBAS integrity monitoring algorithms and hardware configurations to lower the system cost and complexity while maintaining an acceptable safety level [1]. The performance and integrity of LAD-GNSS can also be improved by designing additional onboard modules. The LAD-GNSS integrity monitor testbed was installed on the KAIST campus. However, the above-mentioned modifications, including ground hardware/software changes and airborne components, have not yet been fully analyzed and implemented from a system-design perspective.

In this study, several key components of the LAD-GNSS architecture were developed, including system operation coverage and requirements, modifications of integrity monitors in both ground and airborne subsystems, and theoretical performance evaluations. The operation requirements, vertical alert limits (VALs), are derived based on the operating environment and uncertainty of navigation. The vertical protection level (VPL) under nominal and malfunction case, are compared to VAL to determine if the system meets the integrity requirements for a given operation.

The LAD-GNSS integrity monitors were developed using the well-established GBAS as a starting point. In this study, we implemented major modifications on the ground subsystems by designing additional onboard modules. First, instead of maintaining the ionospheric geometry screening at ground subsystem, which is computationally expensive due to a computation of ionospheric error for the all-in-view geometry, the simplified airborne geometry screening are utilized. In this onboard module, an error induces by ionospheric anomaly is computed for a single geometry which UAV uses for its positioning. This can lower conservatism and computation burden significantly.

In the current GBAS architecture, separate PL values for three scenarios (i.e., fault-free (H0), single reference-receiver fault (H1), and ephemeris-fault cases) are herein defined based on the predetermined integrity and continuity allocations. These allocations can ideally be dynamically assigned to make the PLs identical for all fault

scenarios. The use of UAV information provides a significant advantage over the conventional GBAS and enables allocations in real time. Since onboard module can know information about the satellite geometry utilized by the UAV for its positioning, it is possible to compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

The remainder of this paper is organized as follows. Section 2 illustrates the LAD-GNSS architecture, including the operation coverage and integrity monitoring modifications. Section 3 explains the operation requirement of LAD-GNSS, and Section 4 details the method of theoretical performance evaluation. Section 5 concludes the paper.

2. LAD-GNSS ARCHITECTURE FOR UAV

The proposed LAD-GNSS architecture for UAV emphasizes low-cost and low complexity for commercial UAV applications with compact installation ground/onboard units while supporting high navigation accuracy and integrity within the coverage airspace.

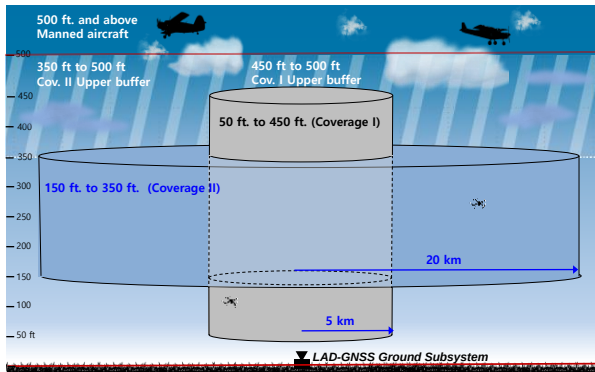


Figure 1. Concept of LAD-GNSS architecture for UAV (no ground obstacle is assumed)

Figure 1 illustrates the operational concept of LAD-GNSS architecture for UAV when there are no ground obstacles within operational regions. The ground facility shown at the center is the source of LAD-GNSS corrections and integrity information as well as real-time guidance for each UAV operating within the coverage airspace. To enable safe operations of UAV in the civil airspace, coverage airspaces were defined around the ground facility. The radius of coverage airspaces of the proposed system was defined based on expected UAV missions for each region. ‘Coverage I’ supports UAV operations within the radius of 5 km from the ground facility. Coverage I is designed to potentially support UAV landing phase down to the low limit of Coverage I with high accuracy/integrity navigation performance. ‘Coverage II’ is designed to support high-speed operations, by having lower navigation accuracy compared to Coverage I, within the radius from 5 to 20 km from the ground facility. The operation range of altitude for each coverage airspace is determined based on considerations of correction accuracy over the distance from the operating

UAVs to the ground facility. Supported altitude for Coverage I is defined from 50 to 450 ft. This indicates that the upper and lower margins for vertical position uncertainty from both upper (500 ft) and lower (0 ft) hard bounds are considered as 50 ft. Supported range of altitude for Coverage II is defined from 150 ft to 350 ft by assuming margins of 150 ft on both upper and lower sides. It should be noted that the upper and lower margins for Coverage II are set larger than those for Coverage I to take into account the degraded correction accuracy when the UAV operates at a longer distance.

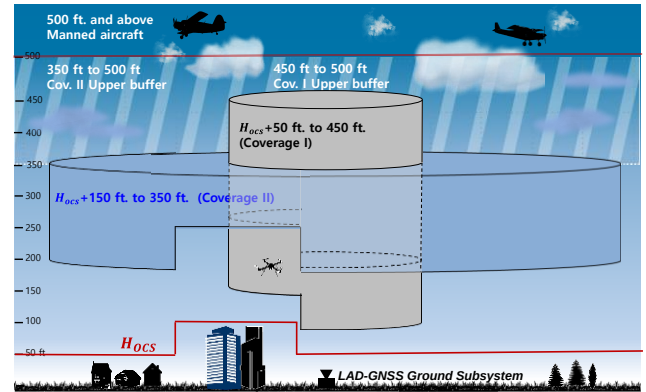


Figure 2. Concept of LAD-GNSS architecture for UAV (when H_{OCS} is assumed)

Table 1. Coverage airspace of LAD-GNSS

Criteria	Radius (from the ground facility)	Altitude
Coverage I	~ 5km	50 ft+ H_{OCS} ~ 450 ft
Coverage II	5km~20km	150 ft+ H_{OCS} ~350 ft

The operational concept introduced in Fig. 1 can be applied in an ideal condition where there are no ground obstacles around operational regions. However, the assumption is not realistic for various commercial UAV operations. Thus, the concept of a height of obstacle clearance surface (H_{OCS}) is introduced. The H_{OCS} indicates the maximum height of an obstacle that can be placed along the aircraft approach path. H_{OCS} for GBAS have already been defined and are enforced at runways that have precision approaches, non-precision approaches, and even visual operations [2]. Since the environments that UAVs fly above is more variable compared to those for manned aircraft, H_{OCS} for UAVs should be adaptively defined based on the environments where the UAV is actually flying above. This H_{OCS} can be pre-defined prior to the operation based on environmental conditions. It is noticeable that the operational concept introduced in Fig. 1 is a special case of the operational concept when H_{OCS} is 0. Figure 2 shows the operation concept with pre-modelled H_{OCS} . H_{OCS} is modeled to bound the heights of buildings and trees in a ground that should not be violated for safety. By assuming the H_{OCS} , the operational range of altitudes

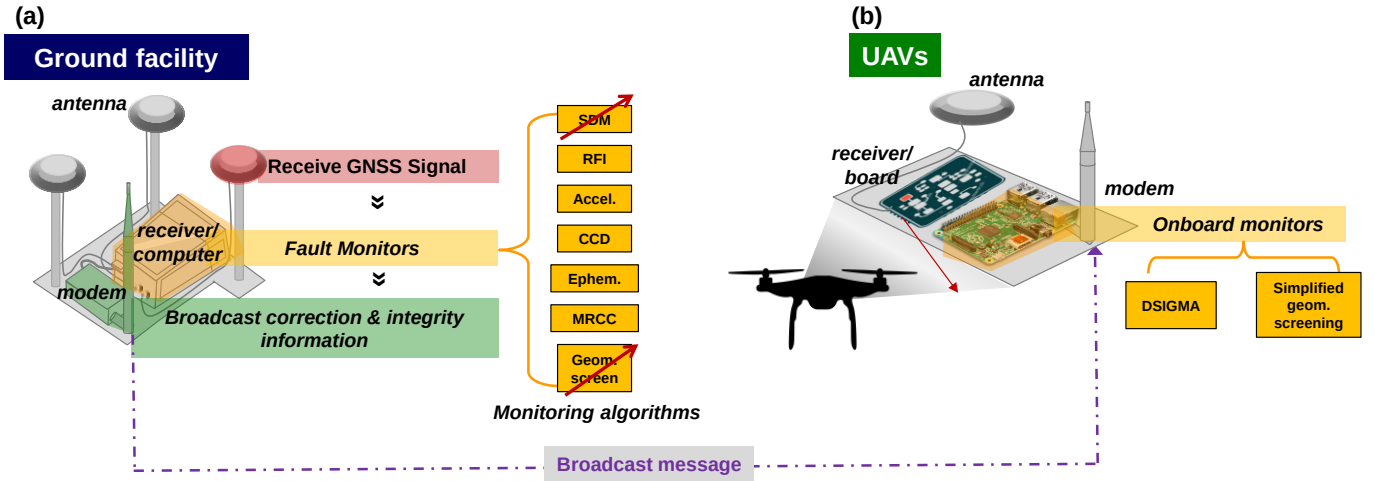


Figure 3. Configuration and components of LAD-GNSS architecture

become variable and reduced. The defined coverage airspaces are summarized in Table 1.

Figure 3 describes the simplified configuration and components of LAD-GNSS that includes the ground facility (Fig. 3(a)) and the UAV onboard module (Fig. 3(b)). Three ground reference receivers are connected to low-cost antennas. By considering siting constraints for most commercial applications will not have room to spread ground antennas over hundreds of meters, antennas are sited close together. Low-quality antennas and relatively short antenna separation increase the impact of multipath errors. To mitigate performance degradation due to the hardware restriction, and to take advantages of the new system configuration, this study develops the several key components of the LAD-GNSS architecture including optimal integrity allocations and the modifications of integrity monitors. The optimal integrity allocations allocate total system integrity risk budget dynamically to make the final position uncertainty bounds identical for all fault modes. This would reduce final user position uncertainty bound compared to the original bound computed from the pre-allocated integrity budgets. The algorithm for optimal integrity allocations is introduced in Section 4.1. In addition to the optimal allocation of integrity budget, modification of monitors can potentially improve the overall performance. To mitigate the integrity risk due to anomalous ionosphere gradients, dual smoothing ionosphere gradient monitor algorithm (DSIGMA) (which is adopted for GBAS CAT-II/III) is additionally employed with the use of a code-carrier divergence (CCD) monitor to meet the system requirement. In addition, geometry screening is moved from ground facility to onboard to reduce the conservatism and computational burdens based on known user satellite geometry on onboard. The simplified geometry screening computes an error induced by the undetected ionospheric gradient in real-time for only a single satellite geometry that user is tracking instead of computing impacts for all the possible user geometries. The concept of modified geometry screening and position uncertainty simulations will be discussed in Section 4.2. Lastly, Signal

Deformation Monitor (SDM) also can be eliminated by selecting ground and airborne GNSS receivers that are either the same or very similar. This would cancel out the resulting error when users apply their differential corrections [1].

3. OPERATION REQUIREMENT FOR LAD-GNSS FOR UAV

PLs are compared to alert limit in each position axis of interest to determine if the system meets the integrity and continuity requirements for a given operation. In this paper, the VALs of Coverage I and II are analyzed based on the total system error (TSE), specifically its composition of the sum of the navigation sensor error (NSE) and flight technical error (FTE). It is sufficient to consider only the vertical requirements since they are more difficult to fulfill than the lateral ones.

The key requirement for vertical TSE performance of LAD-GNSS is ensuring actual UAV location should not violate the H_{ocs} or manned aircraft zone. For example, the vertical TSE of a UAV operating in Coverage I must not exceed 50 ft because the low limit of Coverage I is defined as $50 \text{ ft} + H_{ocs}$. Corresponding to these requirements, the analysis is divided into a nominal NSE case, followed by a malfunction NSE case. The nominal case represents that no faults occur (i.e. fault-free condition) or faults detected by monitors in the navigation system (i.e. faulted condition), and the malfunction case represents that an unmonitored fault occurred that caused a large position error.

3.1 Vertical Alert Limits under Nominal NSE Case

In the analysis of the nominal NSE case, an allowable probability of exceeding the limit for the fault-free condition is 10^{-6} ; An allowable probability of exceeding the limit for the faulted condition is 10^{-5} . These allowed risk can be changed depends on system design or application. The paper handles only the fault-free condition for VAL under nominal NSE case ($VAL_{nominal}$) because it has a tighter requirement than that of the faulted condition. To facilitate the analysis, simplified assumptions are used

in this analytical derivation. It is assumed that the relationship between vertical NSE and $VAL_{nominal}$ at the worst geometry is expressed as

$$\sigma_{NSE,vert}(m) = \frac{VAL_{nominal}(m)}{k_{FFMD}}; \quad k_{FFMD} = 5.43 \quad (1)$$

where k_{FFMD} is a multiplier defined to ensure a specified probability of fault-free missed detection for the case of three ground reference receivers. The vertical TSE is normally distributed with a zero mean and standard deviation ($\sigma_{TSE,vert}$) given by

$$\sigma_{TSE,vert} = \sqrt{\sigma_{NSE,vert}^2 + \sigma_{FTE,vert}^2} \quad (2)$$

It is assumed that, if TSE is zero, the UAV height will be low limits (50 ft from H_{OCS} for the Coverage I or 150 ft from H_{OCS} for the Coverage II). Therefore, the probability that the height of UAV will be lower than H_{OCS} is given by

$$P_{risk,Coverage I} = \int_{50 ft}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,vert}} e^{-\frac{x^2}{2\sigma_{TSE,vert}^2}} dx \quad (3)$$

$$P_{risk,Coverage II} = \int_{150 ft}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,vert}} e^{-\frac{x^2}{2\sigma_{TSE,vert}^2}} dx$$

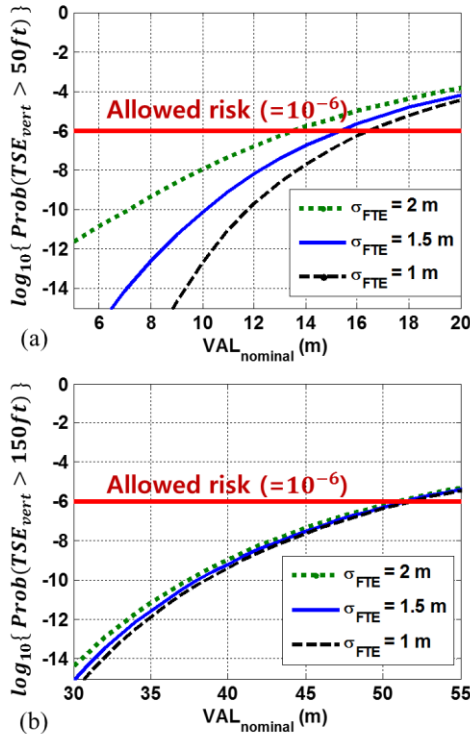


Figure 4. Probability that TSE_{vert} is greater than (a) 50 ft for Coverage I and (b) 150 ft for Coverage II

Figure 4(a) and 4(b) show plots of the fault-free risk versus $VAL_{nominal}$ for values of $\sigma_{FTE,vert} = 1 m, \sigma_{FTE,vert} = 1.5 m,$ and $\sigma_{FTE,vert} = 2m$. Previous works have shown that $\sigma_{FTE,vert}$ of the multi-copter UAV

is a sub-meter level [3]. Based on the conservative value of $\sigma_{FTE,vert} (= 1.5 m)$, the $VAL_{nominal}$ of 15.3 m for Coverage I and the $VAL_{nominal}$ of 51.4 m for Coverage II are determined.

3.2 Vertical Alert Limits under Malfunction NSE Case

An undetected propagation anomalies such as those induced by abnormal ionospheric or tropospheric conditions are considered as malfunctions. These errors resulted from malfunction is assumed as a bias. Therefore, the sum of the bias due to the malfunction and nominal error term must not exceed position uncertainty budgets (50 ft for Coverage I or 150 ft for Coverage II). In the malfunction case, the nominal errors are fixed at their 95th percentiles as shown in Eq. (4).

$$VAL_{malfunction,Coverage I} + TSE_{95\%,vert}(D) \leq 50 ft \quad (4)$$

$$VAL_{malfunction,Coverage II} + TSE_{95\%,vert}(D) \leq 150 ft$$

where D is a distance between the ground facility and the UAV. $TSE_{95\%,vert}$ is given by

$$TSE_{95\%,vert} = 1.96 \sqrt{\sigma_{NSE,vert}^2 + \sigma_{FTE,vert}^2} \quad (5)$$

In this study, the $TSE_{95\%,vert}$ computed by using the $\sigma_{FTE,vert}$ of 1.5 m and $\sigma_{NSE,vert}$ of 1.7 m. Using the Eq. (4) and (5), the $VAL_{malfunction}$ of 10.8 m for Coverage I and $VAL_{malfunction}$ of 41.5 m for Coverage II are determined. As the VALs derived from malfunction case are smaller than the $VAL_{nominal}$ in both coverages, we selected the $VAL_{malfunction}$ as the final VAL for each region.

4. THEORETICAL PERFORMANCE EVALUATION

In the LAD-GNSS for UAV, two VPLs are computed in real time. The first one is VPL for the nominal case that represents a fault-free condition or faulted conditions. By taking benefits from the known satellite geometry, $VPL_{nominal}$ with optimal allocation of integrity and continuity is computed. The other one is VPL for the malfunction that represents the position error due to an undetected ionospheric anomaly. Since the ionospheric anomaly is a representative malfunction case in navigation system error, the only $VPL_{malfunction}$ in ionospheric anomaly case is computed. The resulting VPL_{final} is determined by taking maximum among those two VPLs. When the VPL_{final} is less than or equal determined VAL in Section 3, UAVs operation in each coverage is available.

4.1 Vertical Protection Level under Nominal Case

In the current GBAS architecture, separate PL values for fault-free, single reference-receiver fault, and ephemeris-fault cases are defined based on the predetermined integrity and continuity allocations. For the rest of those fault modes, ground facility ensures that position uncertainty due to other fault mode does not

exceed VPL_{H0} . This concept is commonly known as a maximum-allowable error in range (MERR). The MERR method causes the conservative VPL because they may hold excess integrity allocation. As mentioned in Section 1, using the known satellite geometry of UAV, the onboard module can compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

Figure 5 shows the integrity fault tree of the system to which the optimal allocation was applied. The total integrity risk requirement of 2×10^{-6} per 150 is allocated. As UAV accidents can be expected to have fewer casualties than manned aircraft accidents, the relaxed value of the GBAS CAT I is allocated. In addition, sub-allocations are adjusted by the inclusion of the optimal allocation. In the re-designed integrity fault tree, the integrity risk of single deformation, low signal level, and all other failures are pre-allocated. This is because that the signal deformation error is canceled out when the ground subsystem and UAV use the receivers that are either the same or very similar. Thus, the small value (1×10^{-8} per 150 s) is allocated to the signal deformation integrity risk. In the case of the low signal level integrity risk, the 1×10^{-8} per 150 s is also pre-allocated because the bound of the low signal level is determined based on the impact on the performance of other monitors, not on MERR. Finally, as all other failures are bounded by using the $VPL_{malfunction}$, the integrity risk, which is the same as that of GBAS CAT I, is allocated.

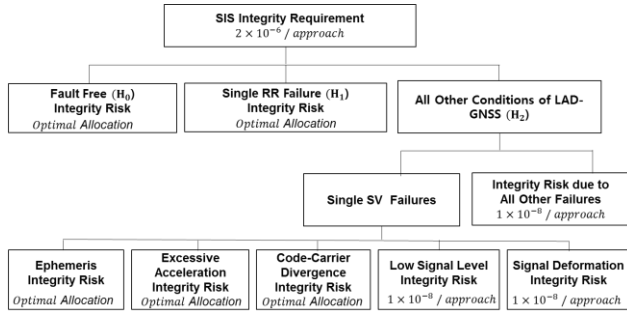


Figure 5. Integrity risk allocation of LAD-GNSS for UAV

As the allocations are optimized with the PLs for each fault mode, the PLs for fault modes that were bounded by using the MERR method should be newly derived. The VPL under the fault hypothesis for satellite i consists of two components, as shown in Eq. (6). First, the position error bound (μ_{fault}) is computed based on the bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{HMI}), which considers that monitors miss-detect such faults. As the μ_{fault} is depended on the fault hypothesis, it is important to find a relationship between the monitor domain errors and user range domain errors to derivate the VPLs for fault mode. In this paper, the VPLs for excessive acceleration (EA) and CCD are developed.

$$VPL_{fault}(i) = \mu_{fault,i} + k_{HMI,EA} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (6)$$

where k_{HMI} is a multiplier defined to ensure a specified probability of the missed detection, and σ_i is the standard deviation of the differential ranging error for satellite i . Additionally, S_{vert} is the row of the weighted-least-squares projection matrix corresponding to the vertical position state [4].

The excessive acceleration threat is a fault condition in a GNSS satellite that causes the range measurements to excessively accelerate. The most probable causes would be GNSS clock anomalies or ground receiver failures by verifying the consistency of the pseudorange and carrier-phase measurements over the last few epochs. The effective differential ranging error ($\delta\rho_{EA}$) due to an acceleration (acc) for satellite i is [5, 6]:

$$\delta\rho_{EA,i} = 0.5 \cdot \tau \cdot (\tau + T) \cdot acc \quad (7)$$

where, latency (τ) represent the time between correction computation from ground subsystem and their application at aircraft side, and T is the measurement update time interval. For the LAD-GNSS ground subsystem, the τ is 1.5 s and T is 0.5 s. Using Eq. (7) and the minimum detectable error (MDE) concept [3], the position error bound (μ_{EA}) for EA is derived as

$$\mu_{EA,i} = S_{vert,i} \cdot 0.5 \cdot \tau \cdot (\tau + T) \cdot (k_{cont,EA} + k_{HMI,EA}) \sigma_{t_{EA}} \quad (8)$$

Here, k_{cont} is a multiplier defined to ensure a specified probability of a fault-free alarm, and $\sigma_{t_{EA}}$ is the standard deviation of the nominal test statistic variation (acc). Once μ_{EA} is derived, VPL_{EA} can be calculated by adding the random error term to μ_{EA} as shown in Eq. (9).

$$VPL_{EA}(i) = \mu_{EA,i} + k_{HMI,EA} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (9)$$

The code-carrier divergence monitor is used to detect ionospheric storms and ensure that the divergence of the code and carrier for any given satellite is sufficiently small. The divergence rate is the estimated change rate of ionosphere delay, I , over time. Given a divergence failure with a CCD rate Div and time onset $t = 0$, a differential ranging error ($\delta\rho_{CCD,i}$) due to a divergence rate for satellite i is [7]:

$$\delta\rho_{CCD,i} = |e_g(t - \max[t_{g,0}, 0]) - e_a(t - \max[t_{a,0}, 0])| \quad (10)$$

where e_g is the ranging error due to Div at ground relative to ground steady state, and e_a is a ranging error due to Div at aircraft relative to ground steady state, $t_{g,0}$, and $t_{a,0}$ is the ground filter start time and airborne filter start time, respectively.

Figure 6 shows the worst-case differential ranging error at Div of 0.004 m/s. Note that prior research [7] suggests that a nominal value of σ_{Div} is 0.004 m/s. The worst case occurs when the ground filter has already converged and the aircraft filter has just begun. For a newly acquired

satellite, the pseudo-range is used after two smoothing time constant intervals, or 200 s, to reach steady state. Therefore, the worst differential range error at σ_{Div} of 0.004 m/s is 0.074, as shown in Fig. 6. Similar to VPL for EA, the VPL for CCD is derived by using the worst differential range error at σ_{Div} of 0.004 m/s and the MDE concept.

$$VPL_{CCD}(i) = S_{vert,i} \cdot (k_{cont,CCD} + k_{HMI,CCD}) \cdot 0.074 + k_{HMI,CCD} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (11)$$

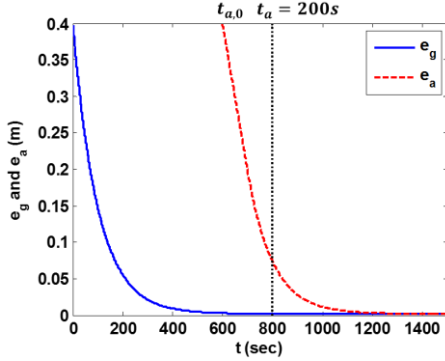


Figure 6. Ranging error due to Div of 0.004 m/s relative to the ground subsystem steady state

4.1.1 VPL with Optimal Integrity and Continuity Allocation

The optimal problem for finding the minima of PLs—which are subject to the constraint that the sum of allocations of each fault mode equals the total risk—is solved by using a Lagrangian function [8]. In this method, all VPLs are expressed as a function of the continuity and integrity risk, as shown in Eq. (12):

$$VPL_i \left(P_{cont,i}, \frac{PHMI_i}{P_{ap,i}} \right) = M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \quad (12)$$

where $P_{cont,i}$ and $PHMI_i$ respectively are continuity and integrity allocated to hypothesis i , and $P_{ap,i}$ is the prior probability of hypothesis i . Once the VPLs are organized into the above expression, a two-dimensional optimization problem can be written as

$$\begin{aligned} & \text{Minimize} \quad \max_i \left[M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \right] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} P_{cont,i} = P_{cont} \\ & \quad \text{and} \quad \sum_{i=0}^{N_{mod}-1} PHMI_i = PHMI \end{aligned} \quad (13)$$

where P_{cont} and $PHMI$ are the total continuity and total integrity risk, respectively, and N_{mod} is the number of hypotheses. Using the results of Section 4.1, the VPLs for LAD-GNSS can be expressed as

$$\begin{aligned} VPL_{H0} &= k_{HMI,H0} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \\ VPL_{H1}(j) &= k_{HMI,H1} \sum_{i=1}^n S_{vert,i}^2 \sigma_{H1,i}^2 + |B_{vert,j}| \\ VPL_e(i) &= k_{cont,H0} \cdot \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{te} \right) \\ & \quad + k_{HMI,H0} \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{te} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{EA}(i) &= k_{cont,EA,i} \cdot (|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{tEA}) \\ & \quad + k_{HMI,H0} \left(|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{tEA} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{CCD}(i) &= k_{cont,H0} \cdot (|S_{vert,i}| \cdot 0.074) \\ & \quad + k_{HMI,H0} \cdot (|S_{vert,i}| \cdot 0.074 + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2) \end{aligned} \quad (14)$$

where σ_{H1} is the standard deviation under a single reference-receiver fault, B_{vert} is the B-value converted to a vertical position error, and x is the displacement vector between the reference station and aircraft antennas. In addition, R_i is the distance (scalar) to satellite i . It is practical at this point to change notations and variables so that the notations are simpler. In Eq. (14), we can change the terms multiplied by continuity, the terms multiplied by integrity, and B-value into θ , ψ , and β , respectively. The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \psi_i + k_{cont,i} \theta_i + \beta_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2a_i Q(-k_{cont,i}) = P_{cont} \\ & \quad \text{and} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (15)$$

where Q is a unit Gaussian cumulative distribution function, a_0 is zero, and $a_i (i \neq 0)$ is one. According to [8], by arbitrarily allocating the continuity budget, the optimal problem can be simplified to a one-dimensional search space. In this suboptimal scheme, parameter $k_{cont,i}$ is therefore no longer a variable, and we can define

$$\beta'_i = k_{cont,i} \theta_i + \beta_i \quad (16)$$

The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \psi_i + \beta'_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (17)$$

In this paper, we implement the suboptimal scheme because the optimal allocation of the continuity offers only a minor improvement while increasing the complexity of the algorithm and computation load [8].

From the suboptimal scheme described above, as well as the VPLs derived in Sections 4.1, simulations of a VPL with an optimal allocation were conducted. For the simulations, the continuity risks were allocated to be the same as those of GBAS CAT-I, and the standard RTCA 24 constellation was used to generate visible satellite geometries. The satellite geometries were simulated every 5 min (which gives 288 epochs for one day).

In addition, we assumed that $\sigma_{t_{EA}}$ is 0.99 cm/s^2 at C/N_0 of 32 dB-HZ and σ_{Div} is 0.004 m/s . For the standard deviation of ground error (σ_{pr_gnd}), the ground accuracy designator (GAD)-A model with $M = 1$ was used. The latter was used because the effect of mitigating the multipath error is not expected in an environment where the separation distances between the antennas are narrow. The standard deviation of airborne error (σ_{air}) was computed by using the airborne accuracy designator (AAD)-B model for the simulation.

Figure 7 shows the resulting VPLs for Coverage I and II when the UAV operated 5 km apart from the ground subsystem with a constant speed of 15 m/s in Coverage I and 20 km apart from ground subsystem with a constant speed of 45 m/s in Coverage II. In this result, the VPLs for fault-free (H_0), single reference-receiver fault (H_1), ephemeris fault cases, EA-fault cases, and CCD-fault cases were computed by using the pre-allocated integrity risk (i. e. without optimal allocations).

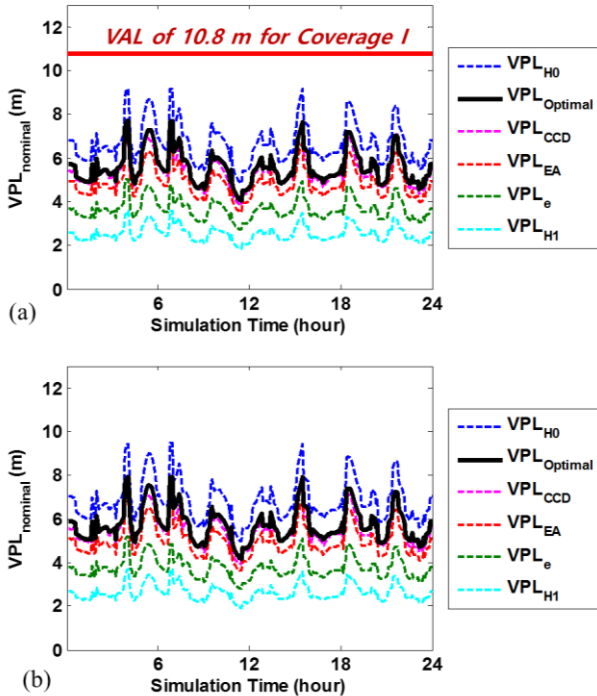


Figure 7. VPLs of LAD-GNSS for UAV operating in (a) Coverage I and (b) Coverage II

As show in Fig. 7, the VPL with the optimal allocation of integrity risk, which is shown by the black curve, is slightly below the VPL_{H_0} . These results show that the VPLs of Coverage I and II are reduced by approximately 16% by using the optimal allocation. The VPL with the

optimal allocation of integrity is reasonably well-bounded by VAL, which is the operation requirement for Coverage I and II.

4.2 Vertical Protection Level under Malfunction Case

To mitigate the integrity risk due to anomalous ionosphere gradients, several monitors including the ground CCD monitor, DSIGMA, and the simplified geometry screening monitor were implemented in LAD-GNSS. The CCD monitor and DSIGMA could detect and exclude the satellite impacted by large ionospheric gradients using the divergence rate of code-carrier measurements. However, the ionosphere gradients that moved at a slow speed were difficult to detect by both the ground CCD monitor and DSIGMA, which resulted in unacceptable large residual errors in GNSS range signals, even after applying differential corrections. Thus, to assess the impacts of anomalous ionosphere gradients on LAD-GNSS users, the worst differential range error was first tabulated using an exhaustive search over combinations of the simulation parameters in the presence of the ground CCD monitor and DSIGMA. Based on the table of the worst differential range errors, $VPL_{malfunction}$ was computed to evaluate the integrity risk of LAD-GNSS users.

4.2.1 Worst-case Differential Range Error Due to Anomalous Ionosphere Gradient

This section describes how we model the worst differential errors in the range domain caused by anomalous ionosphere gradients. In LAD-GNSS, the code-carrier smoothing is processed to mitigate the receiver noise and multipath on both airborne and ground receivers. However, code-carrier divergence due to ionospheric gradients induces bias in the code-carrier smoothing. Such errors in carrier-smoothed code become larger as either a smoothing time or a magnitude of gradient becomes larger. In LAD-GNSS, both the aircraft and ground facility use a time constant of 30 s for the code-carrier smoothing filter in the presence of the anomalous ionosphere gradient.

The resulting differential range errors can be computed by taking the difference between the smoothed ionospheric delay measurements observed from aircraft and the ground facility. Undetected ionosphere-induced differential range errors are then generated for a hypothetical UAV flight in the presence of the modeled ionospheric gradient. The anomalous ionosphere gradient is modeled with several parameters including the magnitude, width, direction, and relative speed of an ionospheric front with respect to a satellite IPP observed from a ground facility. The moving direction of the UAV and the initial position of both the UAV and ionospheric front are also included in the simulation parameters.

The simulation parameters and ranges are listed in Table 2. For a given set of simulation parameters, the differential range errors and the test statistic of the monitors are computed every epoch as the ionospheric front and UAV advance. When the UAV finally reaches

the evaluation point, the differential range error, either detected or undetected, is recorded. Two different UAV operation scenarios for both Coverage I and II are considered in the simulation. The evaluation point is assumed to be 5 and 20 km apart from the ground facility, respectively. The UAV horizontal speed is set at 15 m/s in Coverage I, and 45 m/s in Coverage II, because the speed limit of UAV is constrained to be less than 45 m/s according to the Federal Aviation Administration (FAA) regulations.

Table 2. Simulation parameters

Parameter	Min.	Max.	Step Size
Δv (m/s)	-100	350	10
g (mm/km)	200	500	20
W (km)	25	75	25
α (deg)	-90	90	15
β (deg)	-90	90	10

Figure 8 depicts an example of the results of the worst differential range error in terms of the relative speed of the front. The worst differential range error curves varying with the magnitude of ionosphere gradients from 200 mm/km to 500 mm/km are searched based on the exhaustive simulation over all possible combinations of the ionosphere parameters for each given Δv . The constant model (black line) is proposed to conservatively cover the simulated worst range error curves for all conditions. Since the single-impact scenario is only considered in this analysis, the model of the differential range error does not need to be expressed as a function of Δv . The results of the worst differential range error for the three UAV operation scenarios are summarized in Table 3.

Table 3. Maximum of the worst differential range error for two different UAV operation scenarios

Scenario	Coverage	Horizontal speed of UAV (m/s)	Evaluation Point (km)	Max. range error (m)
(a)	I	15	5	3
(b)	II	45	20	11.5

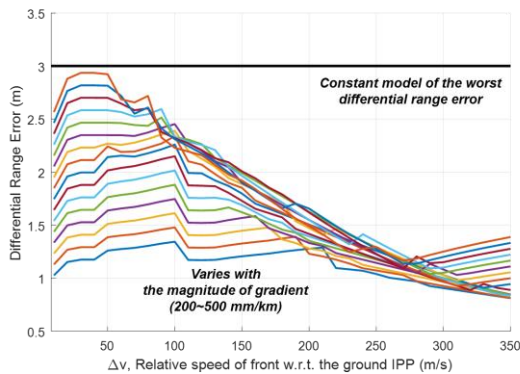


Figure 8. Simulated results of the worst differential range error when the evaluation point is set at 5 km apart from the ground facility and the horizontal speed of UAV is 15 m/s

4.2.2 Vertical Position Error Due to Ionosphere Gradient

The analytical model of the ionosphere-induced differential range error was introduced in Section 4.2.1. Using this model, we examine the largest potential errors resulting from the worst case scenarios. A simulation is conducted to compute the worst-case undetected vertical ionospheric errors (i.e. $VPL_{malfunction}$) by assuming the worst case SV-user geometry and parameter combinations. In this paper, a single-satellite impact is assumed to be the worst case scenario instead of the double-satellite impact mentioned in [9]. This is because the double-impact scenario is a rather conservative assumption for UAV applications.

For a given single satellite i , a vertical ionospheric error is defined as:

$$IEV_i = |S_{vert,i} \varepsilon_i| \quad (18)$$

where $S_{vert,i}$ and ε_i denote the vertical position component of the weighted-least squares projection matrix S and the ionosphere-induced range error for satellite i , respectively. The ionosphere gradient is assumed to move in the worst direction; thus, the resulting Δv is always small. Accordingly, ε_i is always taken as its maximum in the single-impact scenario. Among IEVs computed for all individual satellites in view, the largest IEV is taken as $VPL_{malfunction}$. The final VPL is then determined as the maximum value among $VPL_{malfunction}$ and $VPL_{nominal}$.

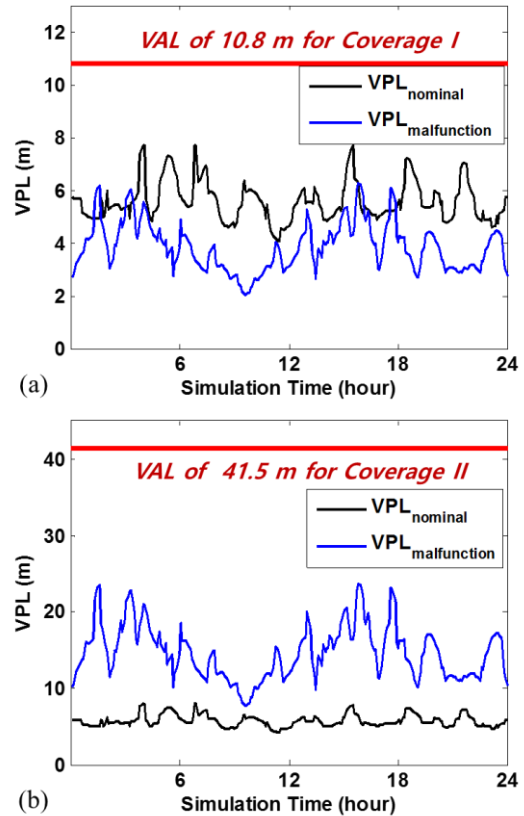


Figure 9. Comparison of the VPL under nominal case, VPL under malfunction case, and VAL for (a) Coverage I and (b) Coverage II

A simulation of $VPL_{malfunction}$ was conducted based on the standard RTCA 24 constellations over the course of a day. For the standard deviation of navigation errors, we used the same model as used in the simulation of VPL under nominal case. $VPL_{malfunction}$ was computed for ideal scenarios. "Ideal" means that the UAV loses no satellite in view. Figure 9(a) shows the result of the $VPL_{malfunction}$ simulation for Coverage I when the UAV operates 5 km apart from the ground facility with a constant speed of 15 m/s. Blue and black curves indicate the $VPL_{malfunction}$ and $VPL_{nominal}$ with the optimal allocation described in Section 4, respectively. The result shows that the final VPLs are determined mainly by $VPL_{nominal}$. On the other hand, in case of Coverage II, the final VPL is determined by $VPL_{malfunction}$ as shown in Fig. 9(b). This is because that the error induced by worst ionospheric gradients sharply increases as the distance between the ground subsystem and the UAV gets larger. As shown in the figure, VPLs for both Coverage I and II were well below the desired alert limits.

5. Conclusion

In this paper, we presented the LAD-GNSS architecture for UAVs that communicate with ground subsystems in real time. LAD-GNSS for UAV was fully analyzed and implemented. It consists of an operation requirement design, integrity monitoring strategies, and theoretical performance evaluation. The operation coverages that were defined for two regions have respective operating requirements. For Coverage I, VPL was compared to VAL of 10.8 m which is determined by analyzing uncertainty of navigation under nominal and malfunction condition. The UAV operating Coverage II was expected to meet the VAL of 41.5 m. In this paper, we show that the final VPL in Coverage I is determined from the $VPL_{nominal}$, while, in Coverage II, the final VPL is determined from the $VPL_{malfunction}$.

In addition, we derived the PLs for EA and CCD fault cases to implement the optimal allocation of integrity/continuity budgets. We simulated the VPL with an optimal allocation. In the simulation results for a 24-satellite GPS constellation, VPLs under nominal case were reduced by approximately 16% when implementing the optimal allocation method.

The NSE and FTE errors highly depended on the hardware, environment, and operational conditions and thus needed to be experimentally evaluated under various conditions. To this end, future work will include both the simulation-based analysis and field tests of several types of UAVs in different flight environments and modes.

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